



Agriculture: Apple Pest Management Guidelines

Phytophthora Root and Crown Rot

Phytophthora spp.

Symptoms and Signs

Symptom expression depends upon how much of the root or crown tissues are affected and how quickly they are destroyed. Generally, crown rots advance rapidly and trees collapse and die soon after the first warm weather of spring. Leaves of such trees [wilt\(/PMG/P/D-AP-PSPP-FS.002.html\)](#), dry, and remain attached to the tree. Chronic infections, usually of the roots, cause reduction in growth and early senescence and leaf fall. These trees may be unthrifty for several years before succumbing to the disease. Phytophthora infections typically kill young trees because their root systems and crown areas are small compared to those of mature trees.

Management

Proper water management is the key to controlling root and crown rot. Do not allow water to accumulate or stand around crowns of trees. Provide adequate drainage or leave unplanted low spots in the orchard, areas that flood frequently, and places where water penetration is poor. Plant on berms. Once *Phytophthora* is present in your orchard, the pathogen will remain; eradication is impossible. Avoid introducing infected plant material, infested irrigation water, or infested soil on farm equipment into uncontaminated soil. Periods of 24 hours or more of saturated soil favor *Phytophthora* infections. Conversely, good soil drainage and more frequent but shorter irrigations (e.g. pulse irrigation) reduce the risk of root and crown rot. Rootstocks vary in susceptibility to the different *Phytophthora* species; none are resistant to all pathogenic species of the fungus. Thus, the success of a rootstock may depend in part upon the species of *Phytophthora* present in the orchard. In general, MM 104 and MM 106 are more susceptible than are M 9 and M 26. Fungicides can help minimize losses.

	Common name	Amount to use	REI‡	PHI‡
	(Example trade name)		(hours)	(days)
Pesticide precautions Protect water Calculate VOCs Protect bees				

	Common name	Amount to use	REI‡	PHI‡
	(Example trade name)		(hours)	(days)
Not all registered pesticides are listed. The following are ranked with the pesticides having the greatest IPM value listed first—the most effective(/agriculture/apple/Fungicide-Efficacy-for-Apple-Diseases/) and least likely to cause resistance are at the top of the table. When choosing a pesticide, consider information relating to the pesticide's properties(/agriculture/apple/General-Properties-of-Fungicides-Used-in-Apples/) and application timing(/agriculture/apple/Treatment-Timings-for-Key-Diseases/), honey bees(/bee-precaution-pesticide-ratings/), and environmental impact(/mitigation/). Always read the label of the product being used.				
A. FOSETYL-AL				
	(Aliette WDG)	2.5–5.0 lb/acre	12	14
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Phosphonate (33)			
	COMMENTS: Foliar spray, 30–60 day interval.			
B. MEFENOXAM				
	(Ridomil Gold SL)	See label; rate varies with method	48	0
		of application and size of tree		
	MODE-OF-ACTION GROUP NAME (NUMBER ¹): Phenylamide (4)			
	COMMENTS: Applications made in early spring and fall.			
‡	Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without protective clothing. Preharvest interval (PHI) is the number of days from treatment to harvest. In some cases the REI exceeds the PHI. The longer of two intervals is the minimum time that must elapse before harvest.			
¹	Group numbers are assigned by the Fungicide Resistance Action Committee (FRAC)(http://www.frac.info/) according to different modes of actions. Fungicides with a different group number are suitable to alternate in a resistance management program. In California, make no more than one application of fungicides with mode-of-action Group numbers 1,4,9,11, or 17 before rotating to a fungicide with a different mode-of-action Group number; for fungicides with other Group numbers, make no more than two consecutive applications before rotating to fungicide with a different mode-of-action Group number.			



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